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REVIEW

Tumor Genomics in Precision Oncology: A Comprehensive Review of Molecular Alterations, Biomarkers, and Translational Therapeutic Advances

Gajavarthini Senthilkumar[1] · **Renuka Saravanan**[1] · **Sivakumar Ramalingam**[1]

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Abstract

Background Cancer is a genomically driven disease characterized by molecular heterogeneity, genomic instability, and therapeutic resistance, necessitating precision-based treatment strategies

Objective This review critically examines recent advances in tumor genomics and their translational applications in precision oncology, with emphasis on emerging technologies, biomarker-guided therapeutics, and adaptive treatment approaches

Methods Relevant literature published between 2015 and 2025 was reviewed using major scientific databases, focusing on tumor genomics, molecular biomarkers, next-generation sequencing, liquid biopsy, multi-omics, artificial intelligence, and clinical translation in oncology.

Results Advances in next-generation sequencing (NGS), liquid biopsy, single-cell and spatial genomics, multi-omics integration, and AI-driven analytics have improved the identification of actionable mutations, predictive biomarkers, and mechanisms of tumor evolution and resistance. These technologies have enhanced patient stratification, targeted therapy selection, immunotherapy responsiveness, and real-time tumor monitoring. Functional genomics and CRISPR-based technologies have further accelerated therapeutic target discovery. However, challenges including tumor heterogeneity, variants of uncertain significance, bioinformatics complexity, ethical concerns, and disparities in access continue to limit clinical implementation.

Conclusion Integrative approaches combining AI, spatial multi-omics, and longitudinal molecular monitoring are expected to advance adaptive precision oncology and enable more dynamic, personalized, and globally accessible cancer care.

Keywords Cancer genomics · Precision oncology · Biomarkers · Tumor heterogeneity · Multi-omics · Immunogenomics · Artificial intelligence

Introduction

Cancer is a multifactorial and genetically driven disease characterized by uncontrolled cell proliferation, invasion of surrounding tissues, and metastasis to distant organs [1]. It arises from the accumulation of genetic and epigenetic alterations that disrupt the balance between oncogenic signaling and tumor-suppressive mechanisms. These changes enable cancer cells to evade apoptosis, sustain proliferative signaling, induce angiogenesis, and acquire invasive and metastatic capabilities, collectively defining cancer as a highly complex and evolving pathological condition [2, 3].

At the molecular level, cancer-associated alterations can be broadly categorized into three major gene classes: oncogenes, tumor suppressor genes, and DNA repair genes [4]. Activation of oncogenes such as *KRAS*, *MYC*, and *BRAF* promotes uncontrolled cell growth, whereas inactivation of tumor suppressor genes, including *TP53*, *RB1*, and *PTEN*, removes critical regulatory checkpoints. In addition, defects in DNA repair genes such as *BRCA1/2* and *MLH1* contribute to genomic instability and increased mutational burden, thereby accelerating tumor progression [5, 6]. Beyond genetic mutations, epigenetic dysregulation, including aberrant DNA methylation, histone modifications, and altered non-coding RNA expression, plays a pivotal role in

✉ Sivakumar Ramalingam
rsiva@src.sastra.edu

1 Department of Chemistry and Biosciences, Srinivasa Ramanujan Centre, SASTRA Deemed to be University, , Kumbakonam 612 001, Tamil Nadu, India

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tumorigenesis by modulating gene expression without altering the DNA sequence [7].

Traditionally, cancer diagnosis and treatment were largely based on histopathological and morphological evaluation, which provided limited insight into the underlying molecular mechanisms [8, 9]. The introduction of techniques such as immunohistochemistry (IHC) and fluorescence in situ hybridization (FISH) marked a significant advancement by enabling the detection of protein expression and chromosomal alterations, thereby laying the foundation for molecularly guided oncology [10]. However, the true paradigm shift occurred with the advent of next-generation sequencing (NGS) technologies, which enabled comprehensive genomic profiling of tumors. Large-scale initiatives such as The Cancer Genome Atlas (TCGA) and the International Cancer Genome Consortium (ICGC) have revealed extensive molecular heterogeneity within and across cancer types, demonstrating that tumors of similar histological origin can exhibit distinct genomic profiles, while tumors from different tissues may share common oncogenic pathways (Fig. 1) [4, 5].

The clinical implications of tumor genomics are particularly evident in the era of precision oncology. Advanced sequencing approaches, including whole-exome sequencing (WES), whole-genome sequencing (WGS), and targeted gene panels, have facilitated the identification of actionable mutations such as *EGFR* mutations in non-small cell lung cancer, *HER2* amplification in breast cancer, and *BRAF* V600E mutations in melanoma [11–13]. These discoveries have enabled the development of targeted therapies and immunotherapies, significantly improving patient outcomes compared to conventional treatments [14]. Furthermore, genomic profiling has expanded the role of biomarkers in cancer management, encompassing diagnostic, prognostic, and predictive applications.

In addition to guiding therapy selection, tumor genomics revolutionized disease monitoring through the emergence of liquid biopsy technologies. Circulating tumor DNA (ctDNA), circulating tumor cells (CTCs), and exosomal biomarkers enable non-invasive, real-time assessment of tumor dynamics, including treatment response, minimal residual disease, and resistance [15]. Despite these advances, significant challenges remain, including tumor heterogeneity, the interpretation of variants of uncertain significance (VUS), integration of multi-omics data, and disparities in access to genomic technologies [16, 17].

Despite the growing number of reviews in tumor genomics, many studies focus either on technological advancements or clinical applications in isolation, with limited integration of multi-omics, artificial intelligence, and real-time adaptive oncology frameworks. This review provides a comprehensive and integrative perspective by bridging molecular mechanisms, emerging technologies, and clinical translation, with a particular emphasis on current challenges and global accessibility. Additionally, it highlights the transition toward adaptive precision oncology, where treatment strategies evolve dynamically based on tumor

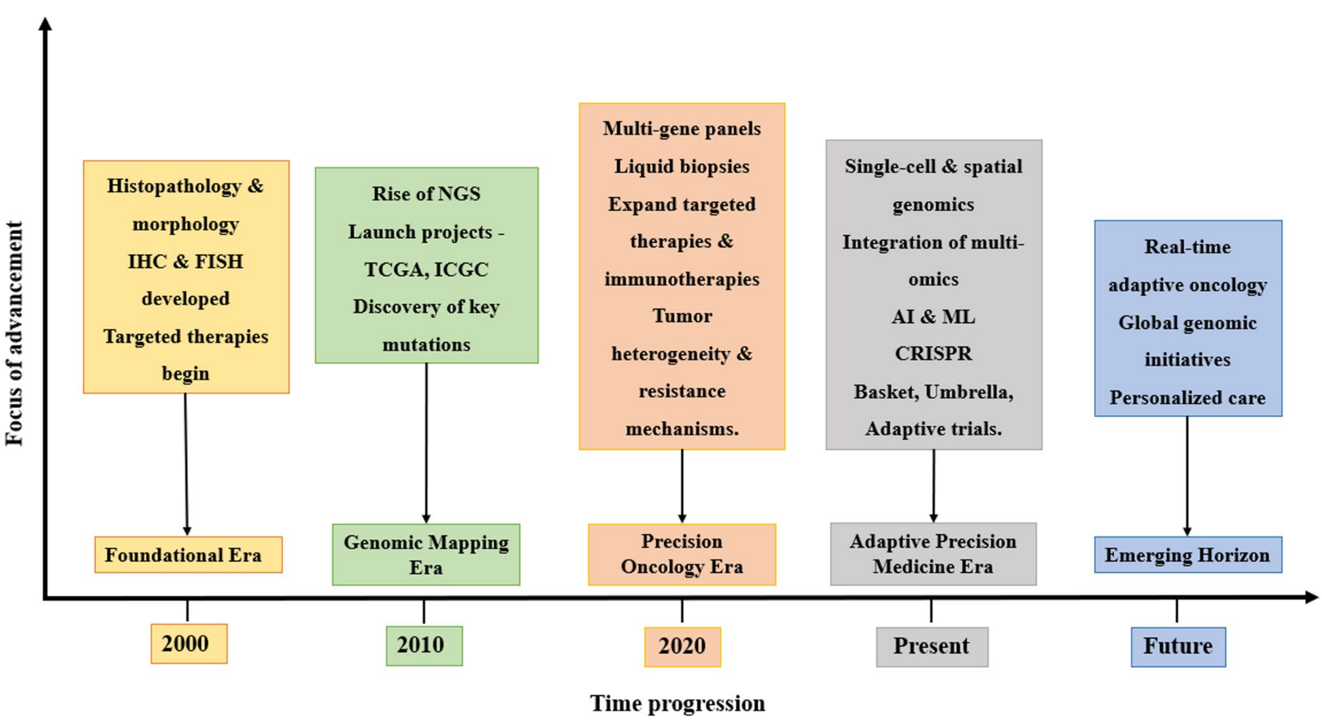

Fig. 1 Timeline of advancements in cancer genomics and precision oncology

genomic profiling, thereby addressing a critical gap between genomic discovery and real-world clinical implementation [18]. Unlike prior reviews that separately discuss genomic technologies or therapeutic applications, this review critically integrates tumor genomics, AI-assisted analytics, spatial multi-omics, immunogenomics, and adaptive oncology frameworks into a unified translational precision oncology perspective [19].

Looking forward, the integration of multi-omics approaches with emerging technologies such as single-cell sequencing, spatial transcriptomics, and artificial intelligence (AI)-driven analytics is expected to refine precision oncology further. These innovations provide deeper insights into tumor heterogeneity, microenvironmental interactions, and evolutionary dynamics, paving the way for adaptive and personalized cancer treatment strategies (Fig. 2) [20, 21]. In this context, tumor genomics has redefined oncology as a data-driven and molecularly guided discipline, necessitating continuous innovation to overcome existing limitations and improve global cancer care.

Materials and Methods

Literature Search Strategy

A comprehensive literature search was conducted to identify relevant studies on tumor genomics and precision oncology. Electronic databases, including PubMed, Scopus, Web of Science, and Google Scholar, were systematically searched for articles published between 2015 and 2025. The search strategy employed a combination of keywords and Boolean operators, including "tumor genomics", "cancer genomics", "precision oncology", "biomarkers", "liquid biopsy", "multi-omics", "artificial intelligence in cancer", and "targeted therapy". In addition, the reference lists of selected articles were manually screened to identify further relevant studies.

Inclusion and Exclusion Criteria

Studies were selected based on predefined inclusion and exclusion criteria. Eligible studies included peer-reviewed

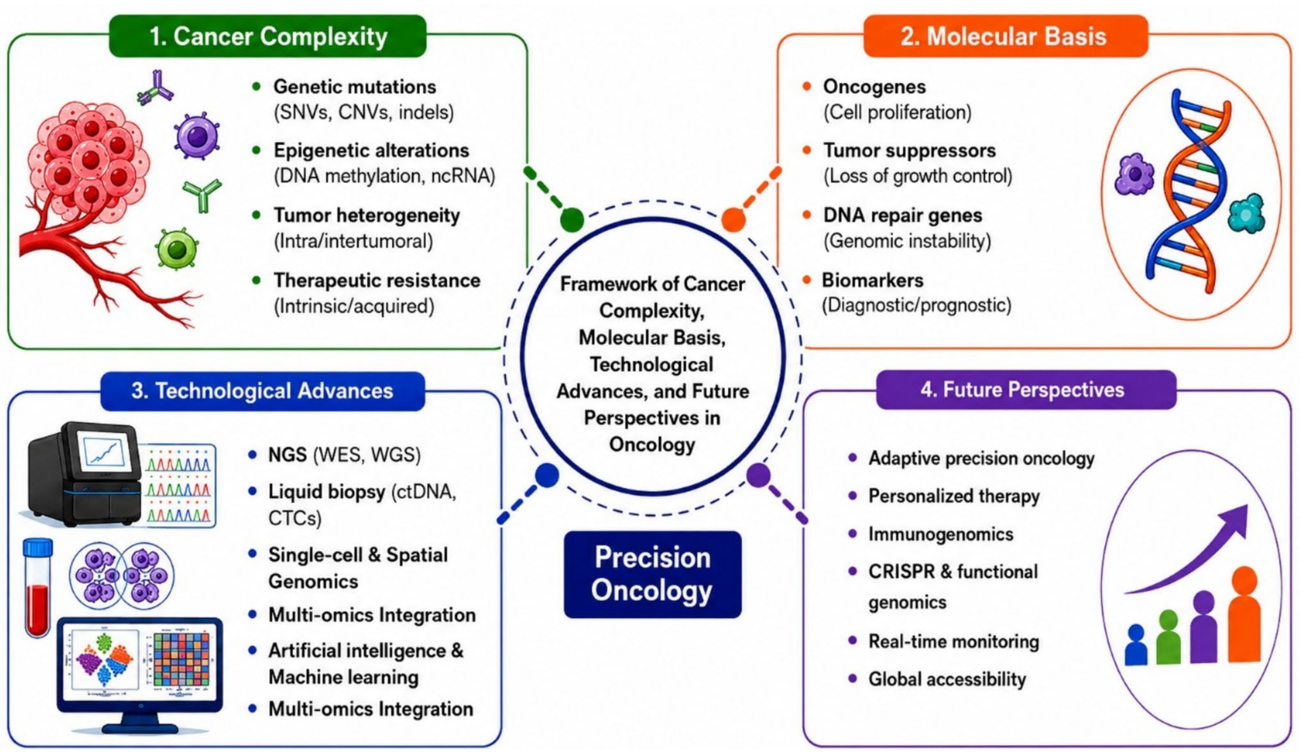

Fig. 2 Framework of Cancer Complexity, Molecular Basis, Technological Advances, and Future Perspectives in Precision Oncology. The figure summarizes the major components driving precision oncology. Cancer complexity includes genetic mutations, epigenetic alterations, tumor heterogeneity, and therapeutic resistance. Molecular basis includes oncogenes, tumor suppressor genes, DNA repair genes, and biomarkers associated with tumor progression and therapeutic response. Technological advances such as next-generation sequencing (NGS), liquid biopsy, single-cell and spatial genomics, multi-omics integration, and artificial intelligence facilitate molecular profiling. Future perspectives focus on adaptive precision oncology, personalized therapy, immunogenomics, CRISPR-based functional genomics, real-time monitoring, and global accessibility for improved cancer care. Abbreviations: SNVs – Single nucleotide variants; CNVs – Copy number variations; ncRNA – Non-coding RNA; NGS – Next-generation sequencing; WES – Whole-exome sequencing; WGS – Whole-genome sequencing; ctDNA – Circulating tumor DNA; CTCs – Circulating tumor cells

research articles and review papers focusing on tumor genomics, molecular oncology, and precision medicine, particularly those reporting clinical applications, biomarkers, or therapeutic strategies. Only articles published in English within the specified time frame (2015–2025) were considered. Studies were excluded if they were conference abstracts, editorials, or non-peer-reviewed publications, lacked relevance to tumor genomics or clinical oncology, contained insufficient methodological details, or were duplicate records.

Study Selection Process

The study selection process involved an initial screening of titles and abstracts to identify potentially relevant articles, followed by full-text evaluation of selected studies. Duplicate records were removed before screening. The selection procedure was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure transparency and reproducibility.

Data Extraction and Synthesis

Data extraction was performed from the included studies, focusing on key parameters such as study design, cancer type, genomic alterations, technologies employed, and reported clinical outcomes. The extracted data were qualitatively synthesized to provide a comprehensive overview of advances, clinical applications, challenges, and future directions in tumor genomics and precision oncology.

PRISMA Flow Diagram

The overall study selection process is summarized using a PRISMA flow diagram (Fig. 3), which outlines the number of records identified, screened, excluded, and included in the final analysis.

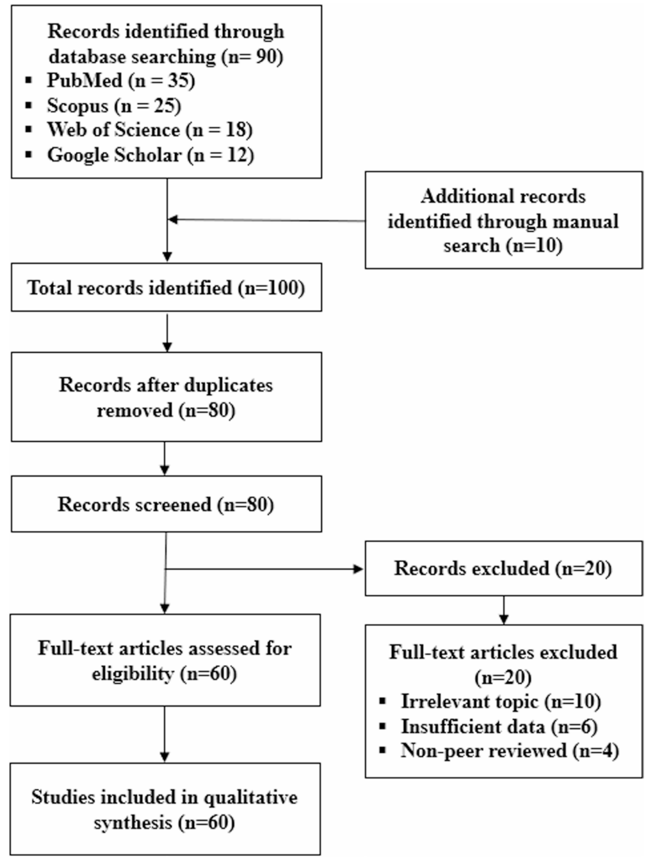

Fig. 3 PRISMA flow diagram illustrating the literature search and study selection process for articles included in this review on tumor genomics and precision oncology

Molecular Basis of Tumor Genomics

Oncogenes, Tumor Suppressors, and Genome Instability

Cancer progression is orchestrated by alterations in three major classes of genes: oncogenes, tumor suppressor genes, and genome maintenance genes, each shaping malignant behavior and therapeutic outcomes. Oncogenes, derived from proto-oncogenes, become constitutively active through mutations, amplifications, or translocations, driving uncontrolled growth [22]. Clinically important examples include *KRAS* mutations in colorectal, pancreatic, and lung cancers (linked to resistance to EGFR inhibitors), *BRAF V600E* mutations in melanoma and colorectal cancer (targetable with BRAF inhibitors), and *MYC* amplifications across diverse malignancies that fuel proliferation and genomic instability. Conversely, tumor suppressor genes act as brakes on proliferation and survival; their loss accelerates tumor development. *TP53* mutations, found in over half of cancers, disable DNA damage checkpoints, while *RB1* loss disrupts G1–S transition in retinoblastoma and small cell lung carcinoma, and *PTEN* inactivation sustains PI3K–AKT signaling in glioblastoma and prostate cancer [23]. A third axis involves genome maintenance genes, where repair defects lead to genomic instability that drives cancer evolution. *BRCA1/2* mutations, central to homologous recombination repair, predispose to breast, ovarian, and prostate cancers and are targeted with PARP inhibitors, whereas mismatch repair defects in *MLH1* or *MSH2* drive microsatellite instability in Lynch syndrome–associated cancers, sensitizing tumors to immune checkpoint inhibitors [24]. Collectively, tumor suppressor loss, oncogene activation, and genome maintenance failure define the genetic basis of malignancy and the actionable vulnerabilities that promote precision oncology. Figure 4 summarizes the major molecular mechanisms involved in tumor genomics, including oncogene activation, tumor suppressor loss, genomic instability, epigenetic alterations, and tumor heterogeneity.

Epigenetic Regulation in Cancer Progression

Epigenetic modifications, which are heritable but reversible changes in gene expression without changing the DNA sequence, play a key role in cancer development by reprogramming transcriptional networks that control genomic stability, growth, and survival. Aberrant DNA methylation, such as CDKN2A promoter hypermethylation, silences key tumor suppressor genes, whereas global hypomethylation promotes chromosomal instability [25–27]. The chromatin structure is reshaped by histone modifications, such as altered acetylation and methylation patterns, which favor tumor suppressor repression or oncogene activation. Non-coding RNAs also contribute, with dysregulated microRNAs suppressing tumor suppressors or enhancing oncogenes, and long non-coding RNAs influencing chromatin remodelling and oncogenic signaling pathways [28, 29]. Crucially, these epigenetic modifications are reversible, in contrast to genetic mutations, which makes them desirable therapeutic targets. Clinically approved agents such as DNA methyltransferase inhibitors (azacitidine, decitabine) and histone deacetylase inhibitors (vorinostat, romidepsin) have demonstrated efficacy in hematological malignancies. Beyond these, emerging therapies are broadening the epigenetic landscape: BET inhibitors (e.g., JQ1, OTX015) disrupt bromodomain proteins that regulate transcriptional elongation of oncogenes like *MYC*; EZH2 inhibitors (e.g., tazemetostat) target aberrant histone methyltransferase activity in lymphomas and solid tumors; and IDH inhibitors (e.g., ivosidenib, enasidenib) reverse oncometabolite-driven epigenetic dysregulation in leukemia [30, 31]. These developments collectively demonstrated the therapeutic potential of epigenetic modification, establishing it as a rapidly emerging field in precision oncology.

Tumor Heterogeneity and Clonal Evolution

Tumor heterogeneity, a characteristic of cancer, exists at several levels, determining disease behavior and response to therapy. Intratumoral heterogeneity is genomic diversity within cancers of the same histology in different individuals. Intratumoral heterogeneity happens when different subclones within the same tumor develop unique genetic or epigenetic changes. As tumors respond to selective pressures like therapy, temporal heterogeneity appears, causing shifts in the clone composition [32–35]. This heterogeneity

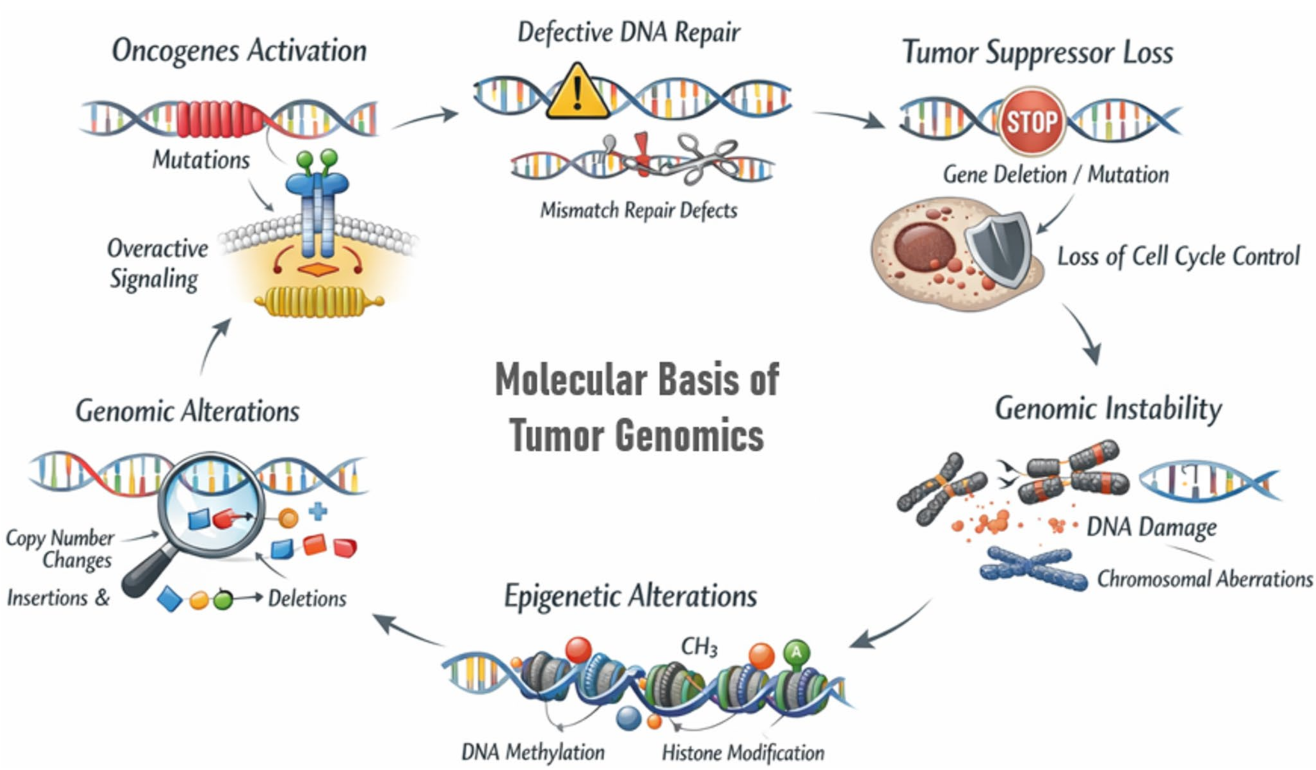

Fig. 4 Composite overview of the molecular basis of tumor genomics illustrating oncogene activation, tumor suppressor loss, genome instability, epigenetic alterations, and tumor heterogeneity

arises from Darwinian clonal evolution, where subclones with beneficial mutations grow and take over the tumor microenvironment, resulting in drug resistance [36]. A well-known example is resistance to EGFR inhibitors in non-small cell lung cancer, often due to the acquisition of the secondary EGFR T790M mutation. Understanding these evolutionary processes is important for developing adaptive treatment strategies, guiding treatment plans using biomarkers, and improving relapse-prevention and disease-monitoring methods.

Technologies Propelling Tumor Genomics

Next-Generation Sequencing

Next-generation sequencing (NGS) has become the cornerstone of precision oncology by enabling high-throughput characterization of tumor genomes and identification of actionable molecular alterations. Whole-exome sequencing (WES), whole-genome sequencing (WGS), and targeted sequencing panels each offer distinct advantages depending on clinical context. While targeted panels provide cost-effective and clinically actionable results, WGS offers broader molecular coverage, including structural rearrangements and non-coding alterations. However, the clinical translation of NGS remains challenging due to inter-platform variability, high computational demands, and difficulties in interpreting variants of uncertain significance (VUS). Furthermore, unequal access to sequencing infrastructure continues to limit widespread implementation, particularly in low- and middle-income settings. Despite these challenges, emerging integration with AI-assisted interpretation and real-time genomic monitoring is expected to improve the clinical utility and scalability of NGS in adaptive precision oncology [37, 38].

Next-generation sequencing technology is the foundation of precision oncology. It has transformed cancer genomics by speeding up the high-throughput examination of tumor DNA. Several strategies are employed based on clinical and research needs. Whole-exome sequencing (WES) covers all the coding regions of the genome and detects most known driver mutations. Targeted gene panels are cost-effective options that focus on genes that have been thoroughly studied. Whole-genome sequencing (WGS) takes a wider approach. It captures both coding and non-coding changes, structural rearrangements, and mutation patterns that offer insights into tumor biology. These sequencing technologies work hand in hand to support therapies guided by biomarkers and tailored treatment strategies. They help identify key changes, discover new treatment targets, and improve molecular-level cancer classification [37, 38].

Transcriptomics, Epigenomics, and Proteogenomics

In addition to changes at the DNA level, multi-omics methods such as transcriptomics, epigenomics, and proteogenomics provide valuable insights into the complexities of cancer. Transcriptomic profiling through RNA sequencing (RNA-seq) enables researchers to measure gene expression, detect alternative splicing junctions, and identify significant gene fusions such as EML4-ALK in non-small cell lung cancer. This influences the use of ALK inhibitors such as crizotinib and alectinib in targeted therapy. Methylation assays are frequently used in clinical diagnostics, including MGMT promoter methylation testing in glioblastoma, to predict a patient's response to temozolomide therapy. Epigenomic profiling studies DNA methylation and histone modifications. It reveals regulatory changes that drive tumor initiation, progression, and resistance to treatment. Proteogenomics offers a practical view of genetic and epigenetic changes by combining proteomic and genomic data. It captures post-translational modifications, protein levels, and signaling pathway activity. For instance, proteogenomic studies have shown drug vulnerabilities in breast and ovarian cancers by highlighting phosphorylation-driven activation of the PI3K-AKT-mTOR signaling pathway. These studies also help improve patient selection for immunotherapies by linking protein-level immune signatures with genomic profiles. Taken together, these approaches provide a multidimensional view of tumors, aiding biomarker discovery, improving molecular classification, and accelerating the translation of biological insights into clinically actionable therapies [37].

Single-Cell and Spatial Genomics

Single-cell genomics allows for a detailed examination of tumor heterogeneity at the cellular level. In contrast, traditional bulk sequencing provides averaged molecular profiles that obscure the variety of cells within tumors. This method identifies rare subclonal populations that can drive tumor growth, spread, or resistance to drugs. It also tracks changes in gene expression and monitors the development of lineages linked to tumor evolution [39]. Clinically, single-cell sequencing has demonstrated how non-small cell lung cancer can resist EGFR inhibitors. This includes the emergence of subclones with the EGFR T790M and C797S mutations. This technique has also been applied in acute myeloid leukemia, helping to identify therapy-resistant stem-like cells that persist after treatment. Additionally, single-cell profiling can show how tumors interact with the immune system, including the exhaustion states of CD8 + T cells. These interactions are key to developing effective immunotherapies [40]. Spatial genomics provides molecular information while keeping tissue structure intact. This helps visualize genomic and transcriptomic changes related to specific tumor types. For instance, this method has highlighted therapy-resistant interactions between stromal and epithelial cells in prostate cancer. It has also identified areas in breast cancer that do not respond to checkpoint blockade. By combining spatial context with cellular diversity, these methods clarify tumor-stroma-immune interactions and the characteristics of the microenvironment. This improves precision oncology beyond standard genomic analysis and opens up possibilities for more personalized, spatially informed treatments [41]. Beyond technological advancements, spatial genomics is increasingly contributing to clinical translation by enabling detailed characterization of tumor–immune–stromal interactions within the tumor microenvironment. Spatial multi-omics approaches can identify immune-excluded niches, resistant cellular ecosystems, and predictive biomarkers associated with immunotherapy response. Although challenges related to cost, computational complexity, and standardization remain, these technologies demonstrate strong potential for real-world implementation in precision oncology by supporting spatially informed therapeutic decision-making [42].

Liquid Biopsy and Circulating Biomarkers

Liquid biopsies provide a less invasive alternative to tissue biopsies. They examine circulating tumor cells (CTCs), circulating tumor DNA (ctDNA), and exosomes released by tumors in blood or other fluids. These tests are helpful in various clinical settings. For instance, ctDNA mutation analysis aids in early cancer detection and confirms diagnoses. They can also identify actionable mutations, such as the T790M EGFR resistance mutation in non-small cell lung cancer (NSCLC). This information supports the use of third-generation tyrosine kinase inhibitors like Osimertinib. Moreover, regular sampling of ctDNA or CTCs allows for real-time monitoring of tumor changes, treatment progress, and minimal residual disease (MRD). This gives insights into clonal adaptation and resistance to therapy. Discovering biomarkers improves by examining exosomal proteins and RNA, which show immune modulation and interactions with the tumor microenvironment. In clinical practice, liquid biopsy has moved from an experimental to a regulatory-approved modality. The FDA has approved ctDNA-based tests, such as the cobas® EGFR Mutation Test v2, to identify EGFR mutations in NSCLC, and commercial multigene panel tests, including FoundationOne Liquid CDx and Guardant360 CDx, are becoming standard for comprehensive genomic assessment in solid tumors, enabling clinicians to identify patients for immunotherapies and targetable therapies. Importantly, these assays are increasingly being integrated into routine care pathways to improve patient stratification, minimize invasive procedures, and facilitate adaptive precision oncology. A clinically relevant example of liquid biopsy-guided adaptive oncology is the monitoring of ESR1 resistance mutations in estrogen receptor-positive (ER+) breast cancer. Circulating tumor DNA (ctDNA)-based detection of ESR1 mutations enables early identification of endocrine resistance and supports timely therapeutic modifications. Emerging artificial intelligence–assisted analytical approaches further enhance interpretation of longitudinal liquid biopsy data, enabling dynamic and personalized treatment adaptation in metastatic breast cancer. They are especially helpful in tumors like lung cancer, where it can be difficult or risky to obtain tissue samples multiple times [43, 44].

Clinical Applications of Tumor Genomics

Genomic Biomarkers in Cancer Diagnosis and Prognosis

Genomic biomarkers have changed how we diagnose and predict cancer. They provide clear molecular information that supports traditional pathology. Some genetic mutations, such as IDH1/2 in gliomas and BCR-ABL in chronic myeloid leukemia, help in classifying diseases accurately. At the same time, high-risk changes, such as TP53 mutations or MYC amplifications, link to aggressive tumor behavior and a poor prognosis. Furthermore, clinically validated gene expression panels, such as the 21-gene Oncotype DX test for breast cancer, stratify patients by recurrence risk and guide treatment intensity. When used, these molecular markers provide tumour-specific fingerprints that improve diagnosis and tailor prognosis and treatment plans to specific cancers [45].

Predictive Biomarkers and Therapy Response

Predictive biomarkers are at the center of personalized therapy by indicating whether a patient has a chance to respond to a treatment. EGFR mutations that render tumors sensitive to tyrosine kinase inhibitors in non-small cell lung cancer, HER2 amplification that foretells how patients with breast cancer will react to trastuzumab, and mismatch repair deficiency or microsatellite instability-high status that foretells how patients will react to immune checkpoint blockade. Above all, secondary resistance mutations, including EGFR T790M or ALK L1196M, underscore their dynamic relevance in shaping future treatment courses and enhancing clinical outcomes [46].

Case-Based Insights: Lung, Breast, Colorectal, and Melanoma

Clinical applications of tumor genomics underscore its leading position in determining therapy orientation in the majority of malignancies. EGFR, ALK, ROS1, and KRAS mutations in lung cancer create genetic subsets treated by targeted therapies, whereas tumor mutational burden (TMB) and PD-L1 expression direct immunotherapy selection. The FLAURA trial, for example, demonstrated that Osimertinib improved progression-free survival in EGFR-mutant non-small cell lung cancer. HER2 amplification is the basis for therapy with trastuzumab in breast cancer, and multigene testing like Oncotype DX and MammaPrint optimizes adjuvant therapy decision-making; the OlympiA trial has already established that patients with high-risk breast cancer and BRCA1/2 mutations will derive benefit from adjuvant PARP inhibitor therapy. In colorectal cancer, KRAS/NRAS wild-type status is predictive of benefit from anti-EGFR monoclonal antibodies, while MSI-H tumors are responsive to immune checkpoint inhibitors. In melanoma, *BRAF V600E* mutations enable BRAF/MEK inhibitor therapy, and high TMB correlates with durable immunotherapy responses. Beyond these, genomic profiling in ovarian cancer has identified *BRCA1/2* mutations that predict PARP inhibitor sensitivity, while in prostate cancer, *AR-V7* splice variants influence outcomes of androgen receptor–targeted therapy. In hematologic malignancies, *FLT3* mutations in acute myeloid leukemia (AML) and *BCR-ABL* in chronic myeloid leukemia (CML) represent paradigm-shifting targets for tyrosine kinase inhibition. Together, these examples show how genomic biomarkers guided by clinical trial data enable personalized, specific treatment strategies across different cancer types [47, 48].

Precision Oncology and Therapeutic Strategies

Targeted Therapies and Resistance Mechanisms

Targeted therapies have changed cancer treatment by taking advantage of new cancer-causing factors, such as the BRAF V600E mutation in melanoma, ALK fusions in non-small cell lung cancer, and PIK3CA mutation in breast cancer. These treatments often yield better results than traditional chemotherapy. However, resistance is unavoidable and can occur through various pathways, including changes in the tumor's characteristics, like the epithelial-to-mesenchymal transition in lung cancer, bypassing pathway activation, or secondary mutations, such as EGFR T790M. The development of next-generation drugs like Osimertinib for EGFR T790M cancers and effective combinations targeting multiple pathways supports the potential of precision oncology. The success of targeted therapy depends on understanding the molecular mechanisms behind resistance [48, 49].

Immunogenomics and Checkpoint Inhibitor Response

Immunogenomics has emerged as a fundamental component of precision oncology by elucidating how tumor-specific genomic alterations influence immune recognition, tumor immune evasion, and therapeutic responsiveness. The success of immune checkpoint inhibitors (ICIs), including programmed cell death protein-1 (PD-1), programmed death-ligand 1 (PD-L1), and cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) inhibitors, has highlighted the importance of integrating genomic biomarkers with immune profiling to optimize patient selection and treatment outcomes.

The Cancer Immunity Cycle provides a mechanistic framework for understanding antitumor immune responses, encompassing tumor antigen release, antigen presentation, T-cell priming and activation, immune cell trafficking, tumor infiltration, recognition of cancer cells, and immune-mediated tumor destruction. Genomic alterations can influence several stages of this cycle by modifying antigenicity, immune visibility, and tumor microenvironment composition. For example, tumors with high tumor mutational burden (TMB), microsatellite instability-high (MSI-H) status, or mismatch repair deficiency (dMMR) often exhibit increased neoantigen generation, making them more responsive to ICIs. Similarly, mutations in DNA polymerase genes such as POLE and POLD1 are associated with hypermutated tumors and improved immunotherapy responses.

However, responses to immunotherapy remain heterogeneous, and resistance mechanisms continue to present major clinical challenges. Genomic alterations such as STK11, KEAP1, and PTEN loss may contribute to immune evasion by creating immunosuppressive tumor microenvironments, reducing T-cell infiltration, or impairing antigen presentation pathways. Importantly, not all patients with favorable biomarkers demonstrate durable responses, highlighting limitations in current predictive frameworks and the need for more robust biomarker combinations.

Emerging approaches integrating immunogenomics with single-cell sequencing, spatial transcriptomics, and artificial intelligence–driven immune profiling are expected to refine patient stratification and improve prediction of therapeutic response. Such integrative strategies may facilitate more adaptive immunotherapy approaches, enable precision-guided treatment selection while minimizing unnecessary toxicity and resistance [49, 50].

Clinical Trial Designs

Basket, adaptive, and umbrella trials are among the novel clinical trial designs that have come to be utilized to address the evolving needs in genomics-informed cancer therapy. NCI-MATCH (Molecular Analysis for Therapy Choice) is a form of Basket trial that evaluates whether a single targeted treatment is effective across multiple tumor types with a shared genetic mutation, such as PIK3CA or BRAF V600E. NCI-MATCH has established regulatory precedent for tissue-agnostic approval, including FDA approval of larotrectinib for NTRK fusion cancers, and has demonstrated that selecting treatment by mutation is feasible. Lung-MAP is an umbrella trial that stratifies patients within a given cancer category into molecularly defined subgroups, each assigned a matched therapy. Lung-MAP has accelerated the development timeline for targeted therapies for squamous non-small cell lung cancer by testing multiple agents at once, maximizing trial efficiency and patient access. Real-time adaptive trials, such as the I-SPY 2 trial in breast cancer, use real-time data to adjust trial arms or patient eligibility, enabling rapid evaluation of multiple investigational drugs. The success of I-SPY 2 has influenced regulatory strategies by showing that adaptive designs can shorten drug development times and improve patient selection. Together, these trial designs illustrate the paradigm shift from organ-specific to mutation-directed oncology, the foundation for the next generation of precision medicine, and in support of regulatory platforms for personalized therapy (Fig. 5) [50].

Challenges in Clinical Translation

Interpretation of Variants of Uncertain Significance (VUS)

One of the challenges of genomic medicine is the common detection of VUS, genetic changes whose impact on tumor biology and on clinical outcome is unknown. Because it is hard to establish their clinical significance, the extensive presence of VUS complicates therapeutic decision-making. This issue is particularly acute in underrepresented groups, in which higher VUS diagnosis rates are due to insufficient reference genomic information. This highlights the pressing

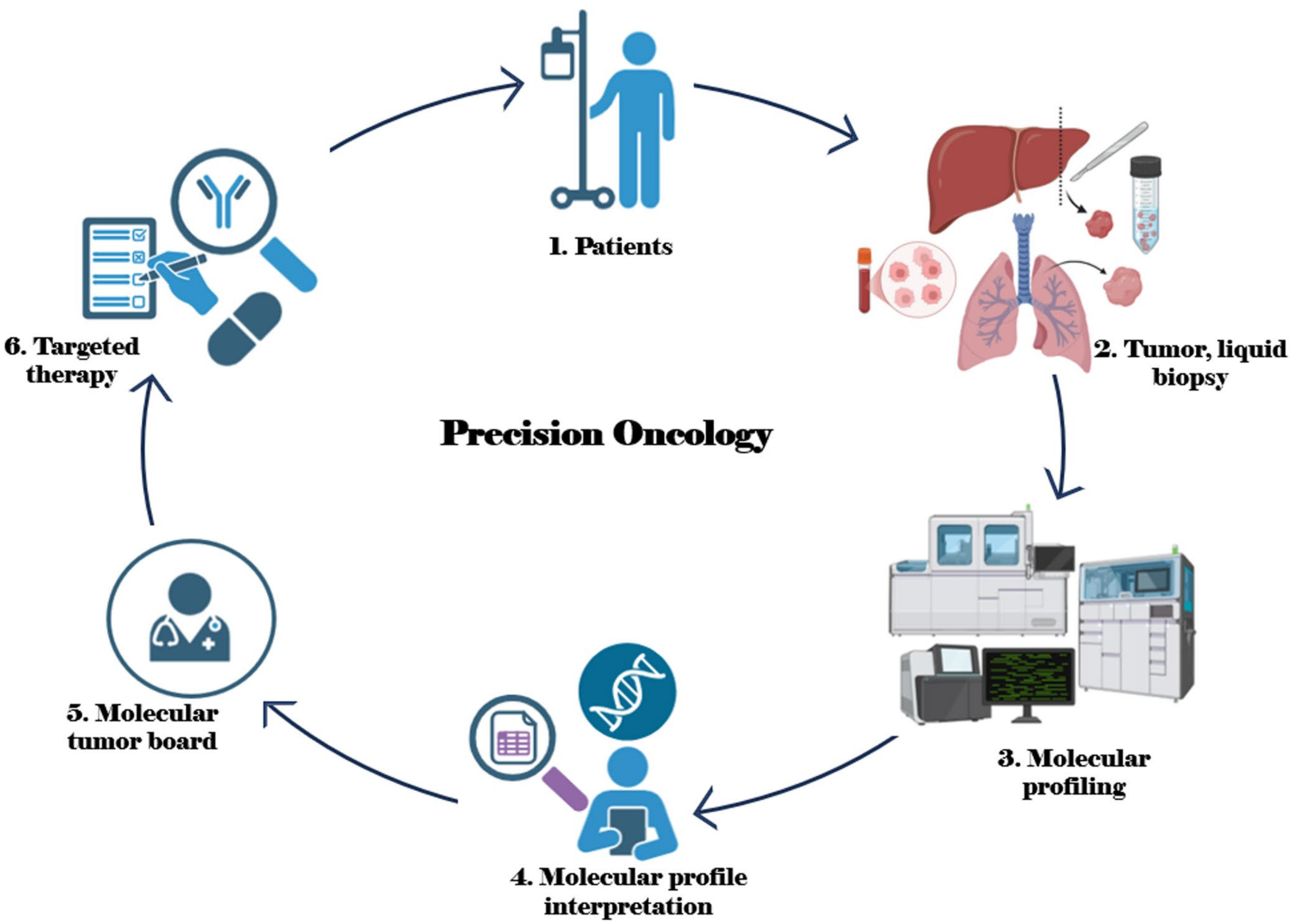

Fig. 5 Clinical workflow of precision oncology

need for large and diverse genetic databases. To overcome this obstacle, integrated frameworks are needed that incorporate large-scale population genomics, functional validation experiments, and sophisticated computational methods to accurately classify VUS, preventing misinterpretation that might lead to unjustified clinical management and ensuring the safe translation of genomic results into precision oncology [51, 52].

Bioinformatics and Data Integration Hurdles

The large volume of data generated by next-generation sequencing (NGS) poses significant analytical and interpretive challenges in precision oncology. Successful integration of genomic, transcriptomic, proteomic, and clinical data requires sophisticated bioinformatics pipelines, solid data standards, and interoperable databases. Clinical interpretation is often complicated by variability in data from reporting formats, annotation methods, and sequencing platforms. In addition, most healthcare environments still lack the specialised cross-disciplinary competency to convert difficult molecular data into actionable clinical insights. These skills are clinical oncology, molecular pathology, and bioinformatics. Meeting these challenges is essential to providing reliable, reproducible, and clinically useful genomic analyses that can be incorporated into daily clinical care [53].

Cost, Access, and Disparities in Genomic Medicine

Whole-genome and targeted genomic profiling remain cost-intensive and unaffordable for most patients, particularly in low- and middle-income nations, despite reductions in the cost of sequencing technologies. Variability in insurance coverage and reimbursement policies further restricts the equitable incorporation of precision oncology into standard practice. These limitations are compounded by socioeconomic disparity, which restricts access to targeted treatments, clinical trial participation, and genomic analysis. To address this problem and ensure that the benefits of genetic medicine are accessible to a wide range of populations worldwide, we need to develop cost-effective diagnostic platforms, invest in healthcare infrastructure, and establish international collaborations to facilitate equitable access [54, 55].

Ethical and Legal Issues

Outside of the clinical environment, genomic testing in oncology raises complicated ethical and legal issues. Protecting patient confidentiality, determining data ownership, handling incidental findings, and avoiding genetic discrimination are significant concerns. The sophistication of informed consent is increasing, particularly as genetic information uncovers inherited risks that affect relatives. While significant safeguards are offered through regulatory systems such as the Genetic Information Non-discrimination Act (GINA) in the US and the General Data Protection Regulation (GDPR) in the EU, significant gaps remain in worldwide governance, policy harmonization, and equal enforcement. These challenges must be addressed through harmonized laws, adequate data protection policies, and effective communication practices in order to ensure ethical use of genomic testing in precision oncology [56].

Emerging Frontiers and Future Directions

Artificial Intelligence and Machine Learning in Oncology

Cancer research is being transformed by advanced analysis and interpretation of sophisticated genetic and clinical data enabled by machine learning (ML) and artificial intelligence (AI). These technologies enable clinical decision-making, digital pathology, variation classification, and therapy response prediction. For instance, deep learning models can extract molecular signatures from medical images and link radiogenomics with treatment response. AI platforms make it easier to apply genetic discoveries to precision cancer treatment, and they improve efficiency by automating data integration and alleviating clinician burden. Precision oncology and artificial intelligence can revolutionize treatment decision-making, prognosis, and diagnosis of various cancers [57–59].

Integration of Multi-Omics for Holistic Patient Profiling

Multi-omics approaches integrate genomics, transcriptomics, epigenomics, proteomics, and metabolomics to provide a comprehensive and systems-level understanding of tumor biology. Unlike single-layer genomic analyses, integrated multi-omics profiling captures the complex molecular interactions that drive tumor initiation, progression, therapeutic resistance, and metastasis. This multidimensional approach enables the identification of combined biomarkers, improves molecular classification, and supports the development of personalized combination therapies. Transcriptomic profiling provides insights into gene expression patterns and signaling pathways, whereas epigenomic analyses reveal regulatory mechanisms such as DNA methylation and histone modifications that contribute to tumor progression. Proteogenomics further bridges genomic alterations with functional protein activity, offering clinically relevant insights into oncogenic signaling networks and therapeutic vulnerabilities. Similarly, metabolomic profiling helps characterize metabolic reprogramming associated with cancer progression and treatment resistance.

Large-scale multi-omics initiatives, including The Cancer Genome Atlas (TCGA) and the International Cancer Genome Consortium (ICGC), have significantly accelerated the translation of integrated molecular datasets into precision oncology. These resources facilitate biomarker discovery, prediction of therapeutic response, and identification of resistance mechanisms, ultimately supporting more precise and individualized cancer management strategies. [57–58]

Functional Genomics and CRISPR Applications

Functional genomics connects genetic changes to their clinical and biological significance. CRISPR-Cas9 genome editing is now a vital tool in cancer research. Pooled CRISPR screens allow for an in-depth study of gene functions [57]. These screens have revealed new oncogenes, tumor suppressors, and synthetic lethal pairs that could be targeted for treatment. For instance, CRISPR screens in triple-negative breast cancer models have shown how BRCA1 interacts with PARP1. This finding paves the way for targeted therapies based on synthetic lethality. Furthermore, homologous recombination deficiency (HRD), initially established in breast and ovarian cancers, is increasingly recognized across other malignancies including prostate, pancreatic, and melanoma. HRD-associated genomic signatures have expanded the therapeutic scope of PARP inhibitors and synthetic lethality–based strategies, highlighting broader opportunities for genomics-guided precision therapy in less extensively studied tumors. In colorectal cancer, CRISPR research has clarified resistance to EGFR inhibitors and highlighted options for combination treatments. Beyond early research, CRISPR technologies are advancing into clinical development. Trials like NCT03872479 are testing CRISPR-edited T cells for stubborn cancers, while NCT04525705 is looking into CRISPR-engineered CAR-T cell therapies for solid tumors. These developments signal a new era in customized cancer treatment, where a precise understanding of genomic changes influences the creation of targeted and adaptable treatment options [59].

Towards Real-Time, Adaptive Precision Oncology

The future of oncology is headed toward adaptive precision oncology, where treatment plans are continuously updated based on real-time molecular and clinical data. Liquid biopsies, such as ctDNA and exosomal RNA, allow for constant, non-invasive tracking of tumor evolution, minimal residual disease, and emerging treatment resistance. For example, the DYNAMIC trial (NCT03786259) is evaluating whether ctDNA-directed decisions on adjuvant therapy in colorectal cancer are effective, and the NCT04585446 trial is looking at ctDNA-directed monitoring to inform therapy in advanced lung cancer. With artificial intelligence–fueled analytics and high-speed sequencing technology added to them, such as Tempus or Foundation Medicine's platforms, these methods allow clinicians to be proactive in therapy adjustment based on the evolving genomic status of each patient's tumor. A new day, where the treatment of cancer keeps pace with the biology of tumors, will dawn with this transition to iterative data-driven care. This approach will improve clinical outcomes, prolong survival, and reduce undue treatment-related toxicity [60].

Roadmap for Future Genomic-Driven Cancer Care

While there has been significant progress, challenges related to data interpretation, access, and ethical use still exist. Tackling these problems will require teamwork across different fields, proper rules, and fair healthcare policies. The combination of AI, multi-omics, functional genomics, and real-time monitoring will improve the accuracy and responsiveness of cancer treatment. Ultimately, these advancements will lead to a future in which genomic-guided oncology becomes the standard of care for all patients worldwide. For example, the UK 100,000 Genomes Project has already analyzed over 16,000 tumor samples. This project combines genomic data and real outcomes into clinicians' decision-making. In Australia, Genomics Australia aims to bring genomic medicine into daily clinical practice, starting with precise cancer treatment. In the United States, the All of Us initiative is gathering genetic and health information from diverse groups to enable personalized prevention, treatment, and care. These breakthroughs highlight global efforts to improve cancer treatment through genomics, ensuring that precision and fair care become the new standard for all patients [60].

Conclusion

Tumor genomics has fundamentally transformed the landscape of cancer research and clinical oncology by enabling a shift from conventional, morphology-based approaches to precision medicine driven by molecular profiling. Advances in genomic technologies, including next-generation sequencing, liquid biopsy, and multi-omics integration, have significantly improved the identification of actionable mutations, biomarker discovery, and patient stratification, ultimately enhancing therapeutic outcomes through targeted therapies and immunotherapies. Despite these advancements, several challenges continue to limit the full clinical implementation of genomics-driven oncology. Tumor heterogeneity, the emergence of therapeutic resistance, and the interpretation of variants of uncertain significance remain major obstacles in translating genomic data into effective clinical decisions. In addition, issues related to data integration, bioinformatics complexity, cost, and unequal access to advanced genomic technologies pose significant barriers, particularly in low- and middle-income countries. Emerging approaches, including single-cell and spatial genomics, artificial intelligence–driven analytics, and functional genomics tools such as CRISPR, are poised to address these challenges by providing deeper insights into tumor biology and enabling more precise and adaptive treatment strategies. The integration of real-time molecular monitoring with advanced computational frameworks is expected to facilitate a transition toward adaptive precision oncology, where therapeutic decisions evolve dynamically in response to tumor progression and molecular changes. In conclusion, while tumor genomics has already reshaped cancer diagnosis, prognosis, and treatment, its full potential will be realized through continued innovation, interdisciplinary collaboration, and the development of cost-effective and globally accessible solutions. Bridging the gap between genomic discovery and clinical application will be critical to achieving equitable and personalized cancer care worldwide.

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Declarations

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